Genes, adaptation and extinction

9A Genetics and evolution · Lesson 6 of 7

Date _____ Class _____

Today I can

- ☐ I can explain how changes in abiotic and biotic factors can make a species endangered or extinct, using the idea of adaptations.
- ☐ I can describe ways of preserving biodiversity, including nature reserves, breeding programmes and gene banks.

What happens today

1. Do Now: the day the dinosaurs died
2. Learn: Adaptations, genes and change · Grey squirrels and red squirrels · and more
3. Core tasks
4. Challenge, then the exit ticket

Student Book pages 16 to 17 · Workbook pages 11 to 12

What I learned today (one or two sentences)

Key Vocabulary

Term	Definition
adaptation	a characteristic that helps an organism survive in its habitat
endangered	a species in danger of becoming extinct
extinct	a species with no living members left
abiotic factor	a non-living part of the environment, e.g. rainfall, temperature
biotic factor	the activities of other organisms, e.g. competition, disease, hunting
competition	when organisms need the same limited resource, e.g. food or light
native species	a species that is naturally found in a part of the world
biodiversity	the number of different species in an area or on Earth
nature reserve	an area where habitats and species are protected
breeding programme	breeding endangered animals or plants in captivity to increase their numbers
gene bank	a store of seeds, gametes or cells kept cold so a species can be grown again

Quick Review

the day the dinosaurs died

About 65 million years ago a giant meteorite hit the Earth near Mexico, throwing up so much dust that it blocked out sunlight. 75% of species became extinct.

1 State one physical environmental factor that the meteorite changed.

2 Explain the effect this had on plants.

3 Explain the effect this had on animals.

Core 1 of 3

Everyone does these. Answer on the card or under it.

1 Match each scientific term to its definition. Draw a line.

environment	a non-living (physical) part of the surroundings, e.g. temperature
environmental factor	the activities of other organisms, e.g. competition, predation
abiotic factor	an organism's surroundings
biotic factor	differences caused by the surroundings
environmental variation	a part of the surroundings that can change an organism

Core 2 of 3

Everyone does these. Answer on the card or under it.

2 Complete the sentences using words from the box. You may use each word once, more than once or not at all.

Word box: abiotic adaptations biotic camouflaged characteristics common desert endangered environmental forest habitats inherited

Animals have _____ to help them survive in their _____ . These adaptations are _____ characteristics. For example, the fur of sand cats matches the colour of the sand in a _____ habitat. The cats are _____ as they hunt gerbils. If more rain fell and the desert got greener, the sand cats would not be _____ and may start to become _____ in that area.

Core 3 of 3

Everyone does these. Answer on the card or under it.

3 Tick one box for each question.

a) What is a native species?

- ☐ A A species that is only active at night.
- ☐ B A species that is eaten by people in a certain part of the world.
- ☐ C A species that is in danger of becoming extinct.
- ☐ D A species that is naturally found in one part of the world.

b) Wolves and foxes compete for the same prey. What type of competition is this?

- ☐ A inter-specific (between species)
- ☐ B intra-specific (within one species)
- ☐ C counter-specific
- ☐ D inner-specific

4 Give three resources that plants compete for.

Challenge 1 of 2

When Core is finished. Full sentences.

C1 About 1500 years ago, giant flightless moa birds lived in New Zealand. They were the prey of huge Haast's eagles. When humans arrived, they hunted the moa to extinction.

a) Explain an adaptation that Haast's eagles might have had for hunting moa.

b) Suggest and explain why Haast's eagles became extinct soon after the moa.

Challenge 2 of 2

When Core is finished. Full sentences.

C2 Explain how a gene bank can allow a plant species to survive if it becomes extinct in the wild, and give one reason why a gene bank alone is not enough to save a species.

Exit Ticket

Hand this card in at the end of the lesson.

Name _____

1 What does it mean if a species is endangered?

2 Give one biotic factor that can cause a species to become extinct.

3 Give one way of preserving biodiversity and say how it works.

Today I can: Goal 1 ☐ Goal 2 ☐